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SMART WOUND MONITORING: AI-DRIVEN IMAGE ANALYSIS BASED ON THE T.I.M.E. FRAMEWORK

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Abstract. Chronic wound assessment is a critical yet subjective process in clinical practice. The assessment is often done based on manual visual inspection, which is very dependent on clinician's experience. The TIME framework evaluates Tissue, Infection/Inflammation, Moisture and Edge conditions, and provides a structured guideline. However, it still remains largely manual and inconsistent. This study proposes TIME-Net, an automated deep learning framework that integrates tissue segmentation with multi-task classification to support more consistent wound assessment. A pre-trained segmentation model, T-SegNet, is used to generate wound masks and produce wound-specific images. These images are then passed into the proposed IME-ClassNet, which simultaneously predicts Infection, Moisture, and Edge conditions. IME-ClassNet applies transfer learning based on EfficientNet-B0 and is enhanced using semi-supervised pseudo-labelling approach to address the limitation of small labelled data. The results showed improvements over the supervised baseline, with Macro F1-score gains of 6.83% for Infection, 83.05% for Moisture, and 12.31% for Edge. These findings demonstrated the effectiveness of semi-supervised learning, specifically for the imbalanced and challenging Moisture task. The final framework was integrated into a smartphone-based workflow using a cloud-hosted inference pipeline, enabling near real-time TIME-based wound assessment. Overall, this study demonstrates the feasibility of combining segmentation, multi-task classification, and semi-supervised learning to support automated wound assessment.

Keywords: Chronic Wound Assessment, TIME framework, Deep Learning, Semi-Supervised Learning, Mobile Integration.

1 Introduction

1.1 Chronic Wound Assessment

Chronic wounds are wounds that do not heal within the expected healing period and often require continuous monitoring to track their healing progress [1]. Some common examples

include diabetic foot ulcers, venous ulcers, arterial ulcers, and pressure ulcers. If these wounds are poorly managed and without any proper dressings, they may lead to delayed healing, infection, hospitalization, and also reduced quality of life [2]. Therefore, a regular wound assessment is essential to help clinicians decide suitable care plans, and also for patients to monitor their wound condition.

Wound assessment is commonly guided by the TIME framework as a standard. The four components in the framework are Tissue, Infection/ Inflammation, Moisture, and Edge. With the full-TIME framework, it provides a structured way to describe the wound condition and support clinical decision-making [3]. However, the TIME-based wound assessment is still commonly performed through manual visual inspection. It is highly dependent on the clinicians' expertise and might be very subjective for different clinicians [4]. At the same time, wound photographs are now widely captured using smartphones for clinical records, remote review, and follow-up monitoring. This creates an opportunity to develop an automated image-based wound assessment system that can support more consistent and accessible wound analysis [5].

1.2 Research Gap

Previous deep learning studies have shown promising results in wound image analysis, especially for wound segmentation and tissue classification. However, many existing approaches focus on isolated tasks, such as detecting the wound area or classifying tissue type. These methods mainly support the Tissue component of the TIME framework, while Infection, Moisture, and Edge are less commonly combined as one workflow. Therefore, this study proposes TIME-Net, a two-stage wound assessment framework that combines T-SegNet for wound-focused image generation and IME-ClassNet for multi-task Infection, Moisture, and Edge classification. The study evaluates the effect of segmentation, compares backbone architectures, applies semi-supervised pseudo-labelling to address limited labelled data, and integrates the final model into a smartphone-based prototype.

2 Related Work

2.1 TIME-Based Wound Assessment

The TIME framework is widely used as a structured approach for wound assessment. It evaluates four key components, namely Tissue, Infection/Inflammation, Moisture, and Edge. These components help clinicians describe the wound condition more systematically and support treatment planning. In clinical practice, the Tissue component focuses on wound bed composition, such as granulation, slough, and necrotic tissue. Infection/Inflammation focuses on signs of bacterial burden or inflammatory response. Moisture refers to the balance of wound exudate, while Edge evaluates the wound margin and healing advancement.

Although the TIME framework provides a structured guideline, its application still depends heavily on manual visual inspection. This can lead to subjective interpretation because different clinicians may describe the same wound differently based on their experience and judgement. Amani et al. showed that the TIME framework is useful for improving wound care competence,

but its practical application still relies on the assessor's ability to interpret visual wound features [3]. Recent reviews on artificial intelligence in wound care also highlighted that many wound assessment tools still focus on limited functions such as ulcer segmentation rather than complete clinical assessment support [4-5]. Therefore, there is a need for a more objective image-based system that can support TIME-based wound assessment.

2.2 Deep Learning for Wound Image Analysis

Deep learning has been increasingly applied in wound image analysis to improve the consistency and efficiency of wound assessment. Existing studies have shown promising performance in tasks such as wound segmentation, wound area detection, and tissue classification. Wound segmentation is commonly used as an early step because it separates the wound region from the surrounding skin and background. This allows the following analysis to focus on wound-related features and reduce the effect of irrelevant background information. Anisuzzaman et al. reviewed image-based artificial intelligence methods in wound assessment and showed that segmentation and classification are two common deep learning tasks in this field [12]. Veredas et al. also demonstrated the use of machine learning techniques for wound image evaluation, showing the potential of computer-aided wound analysis [13].

However, many existing wound image analysis studies focus on isolated tasks. For example, some studies focus mainly on wound segmentation, while others focus on tissue classification or wound measurement. These tasks mainly support the Tissue component of the TIME framework, but they do not fully address Infection, Moisture, and Edge assessment together. This creates a gap for a more integrated model that can use wound-focused images and produce multiple TIME-related outputs in a single workflow.

Transfer learning is also commonly used in medical image analysis because labelled medical datasets are often small. Morid et al. reviewed transfer learning research using ImageNet-pre-trained models and showed that transfer learning is widely used when limited labelled data is available [15]. In this study, EfficientNet-B0 is used as the backbone of IME-ClassNet because it provides a balance between classification performance and model efficiency. The shared feature extractor allows wound-related features to be learned once, while three task-specific heads are used to predict Infection, Moisture, and Edge conditions.

2.3 Semi-Supervised Learning and Mobile Deployment

A major challenge in medical image analysis is the limited availability of labelled data. Manual annotation of medical images usually requires domain expertise, and this makes the labelling process time-consuming. In wound assessment, labels for Infection, Moisture, and Edge may also be subjective because they depend on visual interpretation and clinical judgement. Cheplygina et al. reviewed semi-supervised and transfer learning methods in medical image analysis and highlighted that limited annotated data is a common challenge in the medical imaging field [16].

Semi-supervised learning provides a way to use both labelled and unlabelled data during training. One common method is pseudo-labelling, where a model is first trained using labelled data and then used to generate predicted labels for unlabelled images. High-confidence predictions are retained as pseudo-labels and added to the training set. Liu et al. applied semi-supervised learning with EfficientNet-B0 for chronic wound image assessment by using a small labelled wound dataset together with a larger unlabelled dataset [10]. This approach is relevant to the present study because only a limited subset of wound images has Infection, Moisture, and Edge labels, while the remaining unlabelled images can still be used through pseudo-labelling.

Mobile deployment is another important direction for wound assessment because wound photographs are commonly captured using smartphones for documentation and remote monitoring. Zhang et al. showed that image quality is important for home-based wound monitoring because lighting, angle, and image clarity can affect assessment reliability [14]. Wang et al. also discussed that mobile deep learning applications need to balance accuracy, speed, memory usage, battery consumption, and deployment constraints [11]. Therefore, cloud-hosted inference can be useful for prototype systems because it reduces the need for direct on-device model conversion and allows the model to be updated more easily.

Based on the reviewed studies, there is still a need for an integrated TIME-based wound assessment workflow that combines segmentation, multi-task IME classification, semi-supervised learning, and smartphone-based deployment. This motivates the proposed TIME-Net framework, which uses T-SegNet to generate wound-focused images and IME-ClassNet to classify Infection, Moisture, and Edge conditions in a cloud-integrated mobile workflow.

3 Methodology

3.1 Dataset Description

This study used a publicly available wound image dataset obtained from Kaggle, which contains 2831 wound images compiled from multiple sources [6]-[8]. These images include different wound appearances, lighting conditions, backgrounds, and angles.

This dataset only consists of limited subset that had labels for Infection/Inflammation, Moisture, and Edge based on the TIME framework. These labelled images were used for supervised training and validation of IME-ClassNet, while the remaining unlabelled images were used for semi-supervised pseudo-labelling. As the labelled subset showed class imbalance across IME tasks, especially Moisture, class-weighting loss was applied during model training and Macro F1-score was selected as the main evaluation metric.

3.2 Data Preprocessing

A manual quality check was performed to remove images with severe quality issues, such as extreme blur, incomplete wound visibility, or unusable content. Images captured from different angles but from the same wound were grouped together to prevent data leakage between the training and validation sets.

The remaining images were passed through T-SegNet to generate wound masks. These masks were then overlapped with the original images to produce wound-focused images, where only the wound region remained in RGB colour while the surrounding non-wound area was masked in black. Padding was also added around the wound region to retain the wound edge and nearby periwound area, as these regions may contain useful IME features. The wound-focused images were then resized to 224 x 224 pixels, normalized to the range of 0 to 1, and prepared as the input for IME-ClassNet [9].

3.3 TIME-Net Framework and IME-ClassNet

In the proposed TIME-Net workflow, T-SegNet first generates the wound mask, and the processed output is then passed into IME-ClassNet to predict the IME features.



Fig. 1. Overview of the proposed TIME-Net Framework.

In this study, T-SegNet was treated as a fixed upstream component developed by a collaborator, while this study focused on the development, training, evaluation, and integration of IME-ClassNet.

IME-ClassNet was designed as a multi-task classification model. It uses EfficientNet-B0 as the shared extraction backbone, with pre-trained ImageNet weights applied through transfer learning. The extracted features were passed into the 3 separated classification heads for Infection, Moisture, and Edge. The Infection and Edge heads predict two classes, while the Moisture head predicts three classes. This design allows shared wound features to be learned while producing all IME predictions in a single forward pass.

3.4 Model Training

Supervised Transfer Learning. The training process started with supervised learning using the limited labelled subset. In the first stage, the pre-trained EfficientNet-B0 shared backbone was frozen and only the task specific classification heads were trained. After that, selected top layers of the backbone were unfrozen and fine-tuned with a lower learning rate.

Class Imbalance and Data Augmentation. Data augmentation, including random rotations, flipping, zooms, cropping, and lighting adjustments, was applied to the limited training set to replicate the real-world variations and achieve generalizability. A weighted multi-task loss was used by combining the cross-entropy losses from the three IME task heads.

Semi-Supervised Learning. The semi-supervised learning process was carried out using pseudo-labelling. First, an IME-ClassNet model was trained using only the limited labelled

subset and treated as the baseline model. This baseline model was used to predict labels for the unlabelled wound images and generate a confidence score for each prediction. Only predictions that achieved the pre-set confidence threshold were retained as pseudo-labels. These pseudo-labelled images were added only to the training set, while the validation set remained based on original labelled data to ensure a fair evaluation. This process was repeated across several rounds using different confidence thresholds, and the best-performing model was selected for the next round, until it reached a plateau [10].

3.5 Model Evaluation

This study used a stratified 5-fold cross-validation method to test model performance. Evaluation metrics like precision, sensitivity/recall, and F1-score were used to evaluate the model performance. Macro F1-score was used as the main evaluation metric because the labelled dataset was small and imbalanced. This metric gives equal importance to each class and provides a more balanced view of model performance compared to accuracy alone. The final model was selected based on its validation Macro F1-score across the three IME tasks.

3.6 Mobile Prototype Integration

After model selection, the final IME-ClassNet model was integrated with T-SegNet into a smartphone-based prototype. The model inference workflow was hosted using HuggingFace Spaces. It acts as the cloud inference endpoint, where the uploaded wound image is received and processed by the TIME-Net pipeline. The prediction results from TIME-Net are returned from HuggingFace Space to the Flutter application and displayed to the mobile emulator in Android Studio. The end-to-end response time was measured and recorded for each stage, including tissue segmentation, tissue classification, and IME tasks classification [11].

4 Results

4.1 Effect of Segmentation on IME Classification

A preliminary test was conducted to determine the effect of segmentation on IME classification performance using raw wound images and segmented wound images.

Table 1. Performance comparison with and without segmentation.

Approach	Infection Macro F1	Moisture Macro F1	Edge Macro F1	Average Macro F1
Without Segmentation	0.5564	0.3407	0.5509	0.4827
With Segmentation	0.6929	0.3576	0.7403	0.5969

Table 1 showed the segmented inputs have better IME performance across all 3 tasks. The largest improvement was the Edge classification, followed by Infection classification. Therefore, the segmentation approach was used for the following experiments.

4.2 Backbone Architecture Comparison

There were four architectures compared under the same multi-task classification setting, which were EfficientNet-B0, ResNet50, MobileNet_v3, and Swin_t. In this comparison, the main metrics were Macro F1-score for each task and the model size.

Table 2. Backbone architecture comparison for IME-ClassNet.

Approach	Parameters (M)	Infection Macro F1	Moisture Macro F1	Edge Macro F1	Average Macro F1
EfficientNet-B0	5.3	0.6929	0.3576	0.7403	0.5969
ResNet50	25.6	0.6281	0.4409	0.5913	0.5534
MobileNet_v3	5.4	0.5482	0.3655	0.5653	0.4930
Swin_t	28.3	0.5948	0.4326	0.6304	0.5526

Based on Table 2, EfficientNet-B0 showed the highest average Macro F1-score and the best performance for Infection and Edge classification. Although ResNet50 achieved the highest Moisture Macro F1-score, EfficientNet-B0 had the best average Macro F1-score with fewer parameters. Therefore, EfficientNet-B0 architecture was selected as the backbone for the following experiments.

4.3 Semi-Supervised Learning and Final Model Selection

After that, semi-supervised learning was conducted using pseudo-labelling for 4 rounds. In each round, different confidence thresholds were tested, and the best-performing configuration was selected for the next round. Round 1 to 4 models were first compared using checkpoints selected based on the best total validation loss.

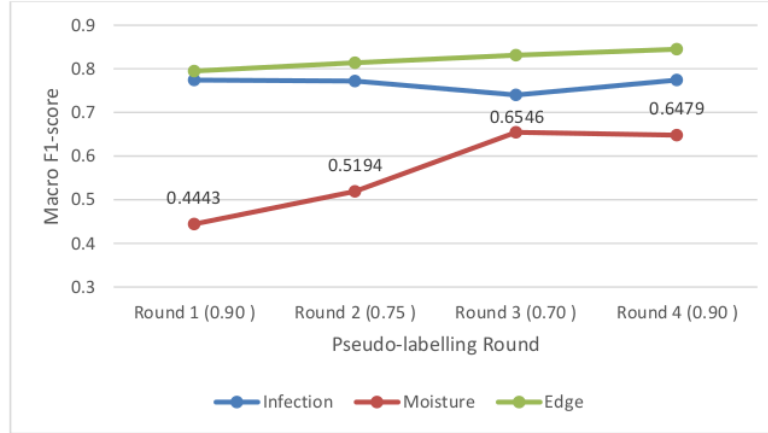


Fig. 2. Macro F1-score across semi-supervised pseudo-labelling rounds using best total validation loss checkpoint selection.

Figure 2 shows that the Edge task achieved the most stable and highest Macro F1-score across all pseudo-labelling rounds. Its performance increased gradually from Round 1 to Round 4. The Infection task showed relatively stable performance, with a slight drop in Round 3 before increasing again in Round 4. In comparison, the Moisture task showed the most noticeable improvement across the pseudo-labelling rounds. The Moisture Macro F1-score increased from 0.4443 in Round 1 to 0.5194 in Round 2 and reached 0.6546 in Round 3. This suggests that pseudo-labelling helped the model learn additional moisture-related visual features from the unlabeled wound images.

However, the Moisture Macro F1-score slightly decreased from 0.6546 in Round 3 to 0.6479 in Round 4. Although Round 4 generated more high-confidence pseudo-labels than Round 3, it did not achieve the best Moisture performance. The active Moisture pseudo-labels increased from 530 in Round 3 to 2243 in Round 4, but the best Moisture Macro F1-score was still achieved by the Round 3 model. This shows that increasing the number of pseudo-labels does not always improve model performance, as pseudo-label quality is more important than pseudo-label quantity.

Table 3. Performance comparison between the supervised baseline and final proposed SSL model.

Task	Supervised Baseline	Proposed SSL Model	Absolute Gain	Relative Improvement (%)
Infection Macro F1-score	0.6929	0.7402	+0.0473	6.83%
Moisture Macro F1-score	0.3576	0.6546	+0.2970	83.05%
Edge Macro F1-score	0.7403	0.8314	+0.0911	12.31%

The comparison in Table 3 showed the final proposed SSL model improved all 3 IME tasks compared with the supervised baseline. The improvement of Moisture task was the largest, where the Macro F1-score increased from 0.3576 to 0.6546, giving a relative improvement of 83.05%. Infection and Edge also improved by 6.83% and 12.31%, respectively. Therefore, the Round 3 Moist_F1 0.70 model was selected as the final proposed model.

4.4 Mobile Prototype Integration

The final proposed model was successfully integrated into a smartphone-based prototype. As shown in Figure 3 (a), the mobile emulator displayed the tissue breakdown result together with the IME assessment. The IME assessment predicted Infection, Moisture, and Edge conditions with their respective confidence scores. The prototype also includes a comparison function for follow-up wound monitoring, as shown in Figure 3 (b), which compares previous and current IME assessment results side by side.

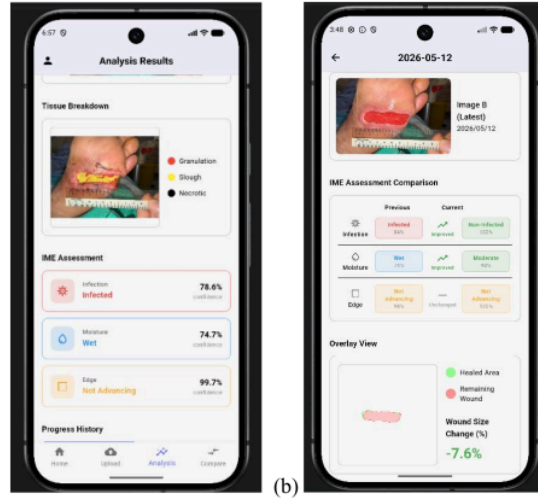


Fig. 3. Mobile prototype interface showing (a) tissue breakdown and IME assessment output and (b) previous and current comparison.

The inference time was also measured to evaluate the feasibility of the cloud-based workflow. The total server inference time was 2692.18ms, which is approximately 2.69 seconds. The segmentation process required the longest time at 1808.94ms, followed by IME classification 473.14ms and tissue classification at 410.09ms.

Table 4. Inference time of the mobile prototype.

Processing Stage	Time (ms)	Time (s)
Segmentation	1808.94	1.81
Tissue Classification	410.09	0.41
IME Classification	473.14	0.47
Total Server Inference	2692.18	2.69

These results show that the proposed TIME-Net workflow can be deployed through a cloud-based inference service and connected to a smartphone application. The prototype was able to return the tissue and IME assessment results within a short response time.

5 Discussion

5.1 Model Selection and Performance

Based on the results, segmented input improved IME classification by reducing irrelevant background information and allows the model to focus on the RGB wound-specific features. This is consistent with Anisuzzaman et al. studies, where segmentation is commonly used to isolate the wound region and reduce background noise before further assessment [12], [13].

The improvement of Infection and Edge tasks was obvious as these two tasks depend strongly on wound boundary and periwound information, such as redness, tissue transition, and wound margin. The padding added around the wound during preprocessing may have helped retain these visual cues. In comparison with Moisture task, the improvement was smaller. This may be caused by Moisture class having 3 different classes (*too_wet*, *moisture*, and *too_dry*), thus, the model needs to distinguish between more categories and increase the difficulty of classification. Zhang et al. developed a mobile health tool for home-based wound monitoring and showed that image quality is an important factor for reliable wound assessment. Their findings help justify that moisture results can be changed depending on lighting, reflection, and exudate visibility [14].

The backbone architecture comparison also supports selecting EfficientNet-B0 as the feature extractor for IME-ClassNet. Morid et al., reviewed transfer learning studies using ImageNet-pretrained models in medical image analysis and proven that transfer learning is commonly utilized when labelled datasets are limited [15]. This matches the data in this study, which is very limited labelled dataset available. Although ResNet50 achieved a higher Moisture Macro F1-score, EfficientNet-B0 achieved the best average Macro F1-score across the 3 IME tasks, while using fewer parameters than ResNet and Swin_t. This supports selecting EfficientNet-B0 as a balanced backbone for overall performance and deployment efficiency.

In Cheplygina et al. study, researchers reviewed semi-supervised and transfer learning methods in medical image analysis and found that limited annotated data is a common challenge in medical imaging [16]. Liu et al. applied semi-supervised learning with EfficientNet-B0 for chronic wound image assessment by using a small labelled wound dataset together with a large unlabelled dataset, which have the similar setup as this study [10]. In their research, they also using unlabelled wound images to improve learning under limited labelled subset. In this study, pseudo-labelling also proved to successfully improve all 3 IME tasks compared with the supervised baseline, with the largest improvement observed in Moisture task. This suggests that the additional pseudo-labelled images increased the model training images and helped the model to learn more visual patterns that were not sufficiently represented in the small labelled subset of the public dataset.

However, the Round 4 result showed that adding more pseudo-labelled data did not always improve the model performance. This may be due to confirmation bias, where the model is confidence in its prediction but incorrect pseudo-labels are added into training and reinforce existing errors. From the results, Round 4 generated many more high-confidence pseudo-labels, but the highest Moisture Macro F1-score was still achieved by Round 3 model. Therefore, the round 3 model was selected as the final model, which was selected based on the validation Macro F1-score rather than the number of pseudo-labels alone.

The mobile prototype integration showed that the proposed TIME-Net workflow can function as an end-to-end proof-of-concept system. The Flutter application was able to display tissue breakdown, IME classification results, confidence scores, and the comparison of wound progressions. Wang et al. discussed that mobile deep learning applications need to balance model accuracy, speed, memory, battery usage, and deployment constraints. In this study, cloud-based deployment reduced the need for direct on-device model conversion and allowed the model to be updated more easily. However, this also means that response time is highly dependent on the user's network speed. Therefore, the current system should be viewed as a prototype that demonstrates technical feasibility, rather than a fully optimized clinical deployment [11].

5.2 Limitations and Future Work

This study has some limitations, which include the small labelled subset, which may limit the model's generalization across different wound types and capture conditions. Although pseudo-labelling increased the training data, but it may still contain errors as it was predicted by the weak model. Second, the IME labels were imbalanced, especially for Moisture task. This limits the model to learn from minority class, even though class weighting and Macro F1-score were used. Third, the model prediction mainly on image color and image visible features. However, some wound conditions may not be observable based on an image and would still require physical wound assessment to confirm their condition. Therefore, the proposed system should be treated as a clinical support tool rather than a replacement for clinical judgement. Future work should include a larger and more diverse dataset, improved pseudo-label filtering strategies, and clinical usability testing. Further deployment work can also explore inference optimization or on-device inference to reduce dependency on cloud connection.

6 Conclusion

This study successfully developed TIME-Net, an AI-driven wound assessment framework that combines T-SegNet for wound-focused image generation and IME-ClassNet for multi-task classification of IME tasks. The findings showed that segmented wound-focused images improved IME classification compared with using original raw wound images directly. EfficientNet-B0 was selected as the model backbone architecture because it is lightweight, which is suitable for mobile deployment and achieved the best overall balance between Macro F1-score and model efficiency. The semi-supervised pseudo-labelling approach also further improved performance compared with the supervised baseline, especially for Moisture task. Round 3 model was selected as the final proposed model as it has achieved the best final Moisture performance. This selected model was also successfully integrated into smartphone prototype through HuggingFace Spaces. This also demonstrated the feasibility of TIME-Net as a wound assessment workflow. Overall, this study shows the potential of combining segmentation, multi-task classification, and semi-supervised learning to support more structured wound assessment. Future work should focus on larger labelled datasets, improved pseudo-label filtering, and deployment optimization before real clinical use.

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Declaration of Generative AI and AI-assisted technologies in the manuscript preparation process. Generative AI tools were used to support language refinement, content organization, and grammar checking. The authors reviewed, verified, and edited all AI-assisted outputs and remain responsible for the accuracy, integrity, and originality of the manuscript.

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